Devi Mutyala

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Education

University of Texas at Austin **2021 – 2026**
Ph.D. in Mechanical Engineering, Oden Institute (GPA: 3.95/4.0) *Austin, TX*

Indian Institute of Technology, Delhi **2019**
M.S. (Research) in Mechanical Engineering (CGPA: 9.22/10) *New Delhi, India*

Indian Institute of Technology, Kanpur **2016**
B.Tech. in Aerospace Engineering (CGPA: 8.2/10) *Kanpur, India*

Technical Skills

AI/ML Frameworks: PyTorch, Hugging Face Transformers, Scikit-Learn, TensorFlow, OpenCV, torchvision
LLMs & Gen AI: OpenAI API (GPT-4, Whisper, TTS), LangChain, Litellm, Ollama, Google Gemma, Meta Llama, Gemini, LoRA / QLoRA, Fine-tuning, Prompt Engineering, Agentic AI, RAG
Data Science Tools: Gradio, BeautifulSoup, SQLite, MLOps (SageMaker), Jupyter, Git, Docker, Requests, lxml
Programming & DB: Python, SQL, ChromaDB, MongoDB, ReactJS, R, MATLAB, Fortran, C
Engineering Software: LAMMPS, COMSOL, ANSYS Fluent, SpaceClaim, PTC Creo, ParaView
Relevant Coursework: Fine Tuning Open Source LLMs, Deep Learning, Multimodal Systems, Machine Learning with Python, Data Science Lab, Software Engineering & Design, LLM Engineering

Professional Experience

Intel Corporation - Data Scientist (Internship) **June 2025 – Dec. 2025**
Design & Patterning AI Group (DPAI) *Santa Clara, CA*

- Engineered interpretable deep-learning defect detection architectures for semiconductor chip layouts; constructed an end-to-end **Computer Vision (CV)** pipeline capable of layout-to-image generation and model training.
- Developed robust ML workflows to address **severely limited labeled data** (~ 100 samples) and **zero non-defect data**.
- Designed and implemented a CNN-based **Prototypical Part Network (ProtoPNet)** to manage representative defect prototypes, optimizing for high coverage across defects. This enabled **explainable AI (XAI)** feature analysis.
- Integrated a modular, **end-to-end ML pipeline** encompassing data augmentation, model training, and results visualization into an existing defect commonality detection codebase.

AI & Machine Learning Projects

Advanced Retrieval-Augmented Generation (RAG) System | *LangChain, OpenAI API, Litellm, ChromaDB, Gradio*

- Architected a sophisticated dual-pipeline RAG system, benchmarking a baseline **LangChain** implementation against a custom-built advanced retrieval engine utilizing **OpenAI SDK** and **Litellm** for superior querying.
- Engineered an advanced ingestion pipeline featuring **semantic document segmentation** and zero-shot **LLM chunk summarization** to radically maximize embedding contextual density within local **ChromaDB** vector storage collections.
- Implemented dynamic **query rewriting**, multi-query concurrent retrieval, and precise **LLM-based reranking**, boosting performance over the **baseline LangChain MRR (0.85) to an exceptional 0.88 MRR**.
- Developed a comprehensive **LLM-as-a-judge evaluation framework** with interactive **Gradio** dashboards, validating improvements from the **92% baseline to a 95.5% keyword coverage rating** and increasing absolute accuracy to **4.51/5**.

Medical Visual Question Answering (VQA) | *Gemma, LoRA, Hugging Face, Gemini, BioCLIP, DenseNet, BERT*

- Built a multimodal Medical Visual Question Answering (VQA) system leveraging the comprehensive **VQA-RAD dataset** to autonomously generate highly detailed clinical explanations and diagnostic insights for complex radiology scans.
- Fine-tuned a generative text model (**Gemma-3-4B-it**) via **QLoRA**, and engineered a custom descriptive dataset using **Gemini-2.0** to augment short-answer inputs for enhanced training convergence (**BLEU 0.23, ROUGE F1: 0.48**).
- Developed high-accuracy benchmark classification pipelines utilizing foundational visual models including **BioCLIP + BERT** alongside custom architectures integrating **DenseNet121 + BiLSTM** for robust baseline validation.

Multimodal Flight Assistant | *Agentic AI, OpenAI Tool Calling, Whisper, PyTorch, SQL*

- Architected a highly scalable **multi-agent system** featuring a **dual-LLM orchestration architecture** to dynamically route open-ended dialogue intent and normalize geographic entities for executing complex flight booking workflows.
- Engineered an **Agentic AI** pipeline securely leveraging **OpenAI Tool Calling** to map conversational user intent precisely to real-time **SQL** database queries, enabling autonomous ticket price retrieval and immediate dynamic UI updates.

- Built a complete end-to-end multi-modal conversational interface seamlessly integrating **OpenAI Whisper (STT)**, generalized **TTS** engines, and **Gradio** frontends to deploy entirely hands-free voice-to-voice interactive application flows.

Smart Audio Scribe | *OpenAI Whisper, Pyannote, Meta Llama, Hugging Face, PyTorch*

- Architected an end-to-end **Speech-to-Text (STT)** integration pipeline comprehensively leveraging **OpenAI Whisper** for high-fidelity transcription alongside **Pyannote** for precise **speaker diarization** of complex multi-speaker audio recordings.
- Engineered a complete **Generative AI** engine operating purely on local **Meta Llama LLMs** via **Hugging Face Transformers** to securely and autonomously parse raw transcripts and synthesize structured **context-aware meeting minutes**.

AI Financial News Engine | *GPT-4o, GPT-5-nano, BeautifulSoup, Python, Requests*

- Engineered a rapid intelligent financial data pipeline utilizing **GPT-5-nano** to continuously and autonomously classify, filter, and extract the most objectively relevant market news from heavily web-scraped (**BeautifulSoup**) Yahoo Finance web page.

Find Common Object (Explainable AI / ProtoPNet) | *PyTorch, torchvision, CNN, Matplotlib*

- Developed a convolutional neural network (**CNN**) **Explainable AI** model implementing Prototypical Part Networks (**ProtoPNet**) to discover, isolate, and precisely localize common sub-objects across visual datasets.
- Designed sophisticated deep learning models that inherently learn human-interpretable image structural prototypes rather than black-box embeddings, visually providing spatial localization strictly through **bounding boxes** and rich **heatmaps**.

Engineering & Research Experience

Oden Institute, University of Texas at Austin – Graduate Researcher **Sept. 2022 – May 2025**

Advisor: Prof. Narayana Aluru *Austin, TX*

- Investigating complex physics of nanoscale systems to explore the development of a **nanofluidic memristor**.
- Employing **Molecular Dynamics simulations (LAMMPS)** and **Poisson-Nernst-Planck continuum models (COMSOL)** to analyze ionic transport and electrokinetic behavior in nanoscale environments.
- Utilizing **Python** to develop analytical frameworks and conduct comprehensive **data analysis**, uncovering insights connecting physical mechanisms with anomalies in large-scale simulation results.

Indian Institute of Technology, Madras – Project Associate **Aug. 2020 – Apr. 2021**

Department of Aerospace Engineering *Chennai, India*

- Executed **Large-Eddy Simulations (LES)** of high Reynolds number channel flows with non-equilibrium effects.
- Implemented **wall-modeled LES (wall stress model)** for non-equilibrium flows and sensor-based computations for accurate transition flow modeling into high-fidelity **in-house CFD solvers**.

Havells India Ltd. – Research Engineer **July 2019 – Aug. 2020**

Center for Research and Innovation *Delhi, India*

- **Patented Venturi Air Cooler:** Designed an energy-efficient device eliminating water pumps by harnessing pressure/temperature profiles. Secondary inventor on patent **IN202011018315**.
- Developed analytical models leveraging **Blade Element theory** and **CFD** to design aerodynamic hardware, significantly improving volumetric flow rates and **minimizing power consumption** before scalable production.

University of Texas at Austin / IIT Delhi – Teaching Assistant **2017 – 2025**

Mechanical Engineering Departments *Austin, TX & Delhi, India*

- Instructed tutorial sessions and conducted automated evaluations using Python for Applied Thermodynamics (2025), Computational Fluid Dynamics (2022), Propulsion (2021), and Gas Dynamics (2018).

Publications & Patents

Publications: *Ozonated Monolayer Graphene for Extended Performance and Durability in Hydrogen Fuel Cell Electric Vehicles.* Devi Mutyala, Deji Akinwande, et al. ACS Nano (2025).

Publications: *Room-Temperature Pulsed Laser Deposition of Boron Nitride for Enhanced Fuel Cell Selectivity.* Devi Mutyala, Deji Akinwande, et al. ACS Nano (2025).

Patents: *Venturi Air Cooler.* Issued January 17, 2024. Patent Number IN202011018315.

Awards & Achievements

Academic Awards: Recipient of the **UT Austin Engineering Fellowship** (2021-2025) and the **IIT Kanpur Academic Excellence Award** awarded to the top 5% of the class (2014-2015).

Leadership and Service: Recipient of the **Institute Blues** (2016), the highest honor at IIT Kanpur for exceptional sports leadership and institutional impact as Team Captain and Deputy Contingent Leader.

Athletics Achievements: Attained 8 Inter IIT Athletics records and 11 Gold medals in track and field at IIT. Served as Captain of the Institute Athletics Team and Deputy Contingent Leader.